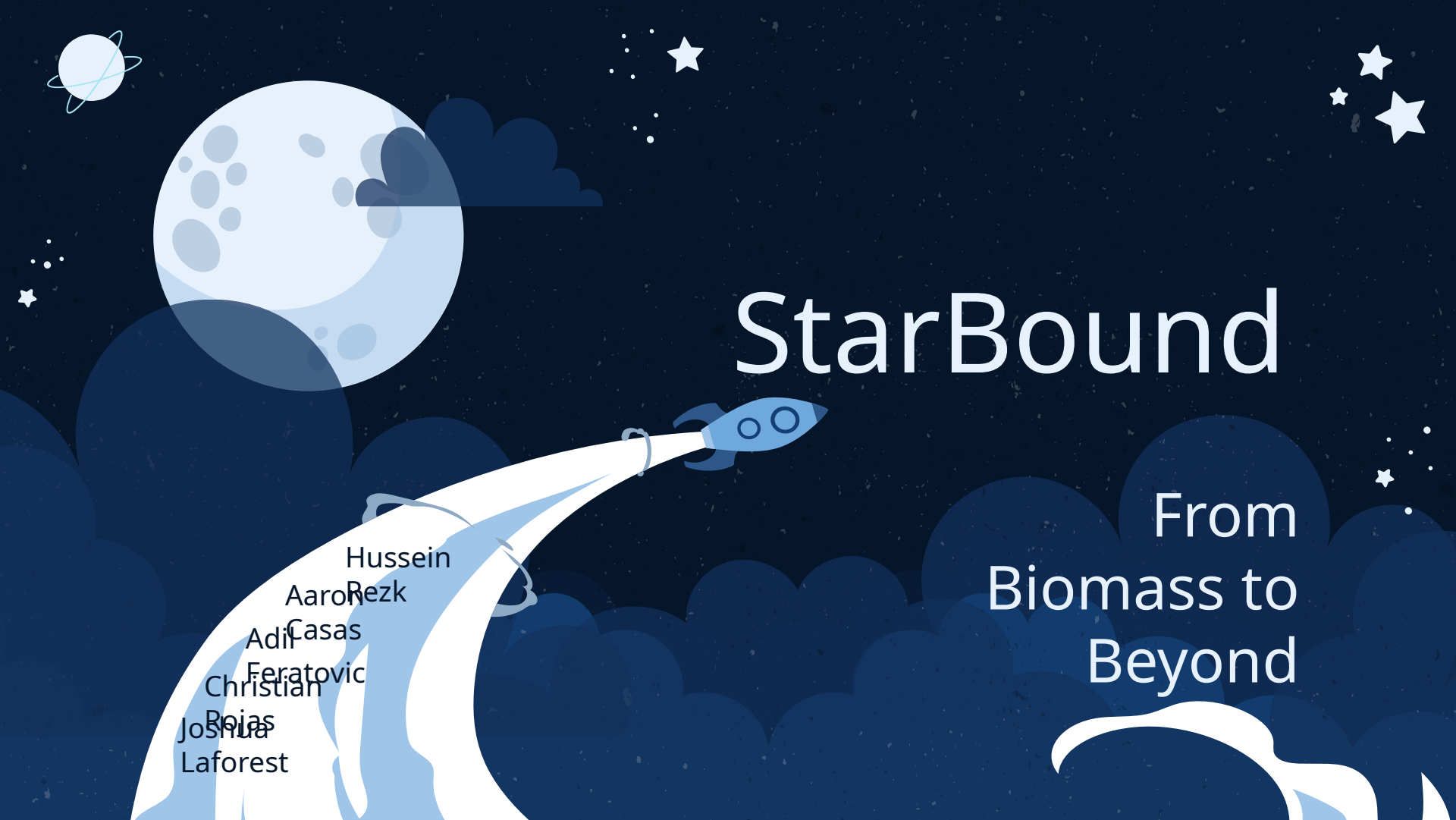


A stylized illustration of a space scene. A blue rocket with two circular windows is launching from the bottom left, leaving a long, white, curved trail that arcs across the frame. In the upper left, a large, light blue moon with darker blue spots is partially obscured by a dark blue cloud. The background is a deep navy blue, speckled with white stars of various sizes and a small planet with a ring in the top left corner. The title 'StarBound' is written in a large, white, sans-serif font in the upper right, and the subtitle 'From Biomass to Beyond' is in a smaller, similar font below it. The names of the team members are listed along the white trail of the rocket.

StarBound

From
Biomass to
Beyond

Hussein
Aaron Rezk
Adil Casas
Christian Feratovic
Joshua Rojas
Laforest

Overview

Introduction



Updated BFD



Updated PFD



Equipment Sizing



Safety and Process Control



Updated Economic Analysis





Introduction



Problem Significance

Space industry is expected to hit **\$1.8 trillion** by 2035

Current RP-1 is made using fossil fuels, high scope 1 emissions

Amazon adopting net-zero policies in their efforts to launch space satellites



What?

Biomass derived RP-1



Where?

California



How?

Biomass→Bio-Oil→RP-1

Market Analysis



Global Growth Insights. (2025, June 18). Rocket Fuel Market Size, share & report [2033]. Rocket Fuel Market Size, Share & Report [2033]. <https://www.globalgrowthinsights.com/market-reports/rocket-fuel-market-115202>

- Space industry is expected to hit \$1.8 trillion by 2035
- Average growth rate of 9% per annum
- Limited supply since there are limited suppliers

Market Analysis

Location	Global Rocket Fuel Demand	Rocket Fuel Used
US	35-40%	LOX/RP-1, LOX/methane
Asia-Pacific	30-35%	LOX/LH2
Europe	20-25%	LOX/LH2
Middle East & Africa	5-10%	LOX/RP-1

- In 2024, the US averaged 60-70 launches weekly
- China launched over 60 missions in 2023

The background is a dark navy blue with a subtle pattern of small white dots, resembling a starry night sky. In the top left corner, there is a dark blue cloud with several white stars of varying sizes. In the top right corner, there is another dark blue cloud. Along the bottom edge, there is a row of overlapping dark blue circles, some of which contain small white stars. The text 'BFD' is centered in the middle of the image in a large, white, sans-serif font.

BFD



3 Stage Process



Fast Pyrolysis

Biomass to Bio-Oil



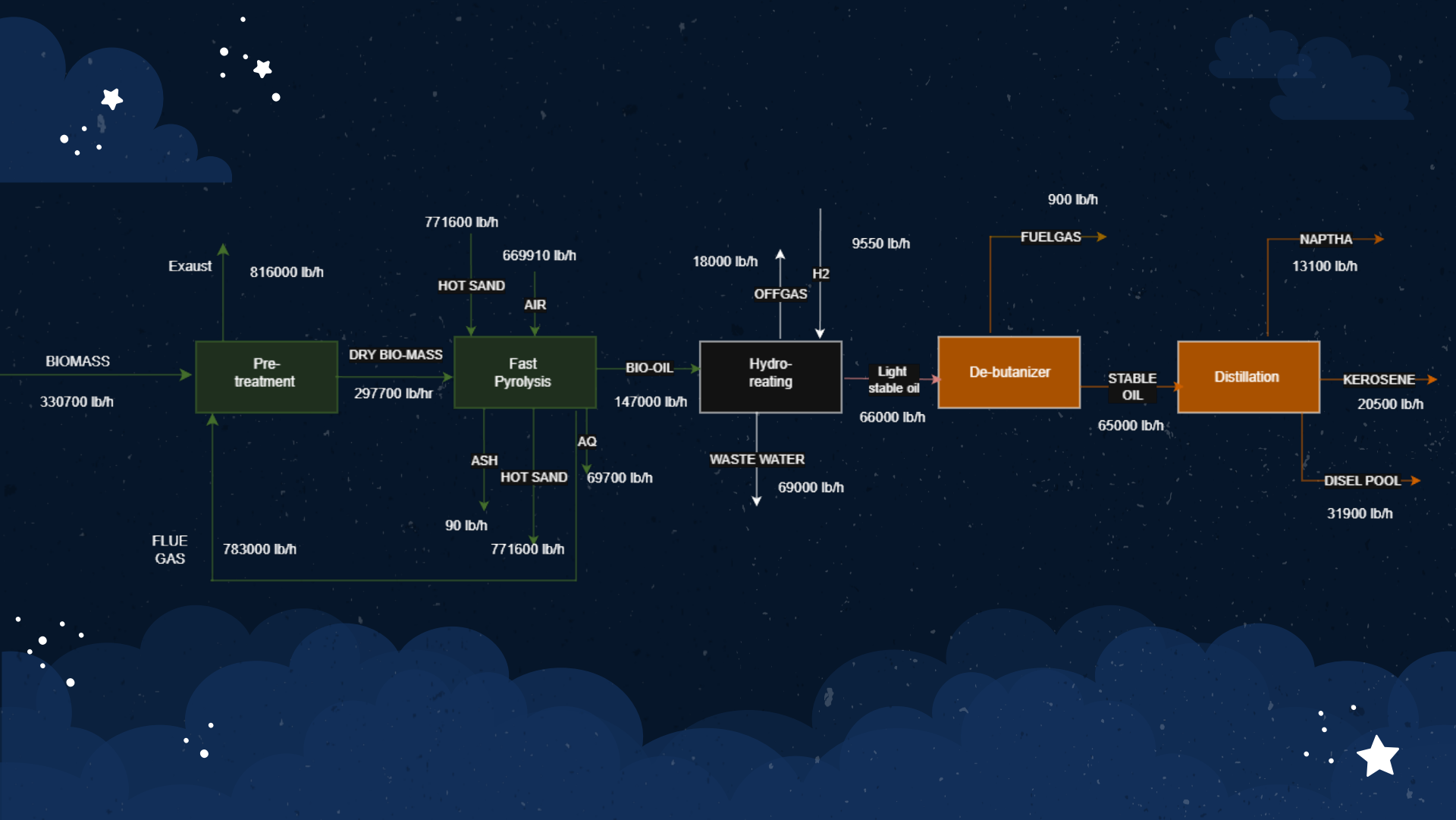
Bio-Oil Upgrading

Stabilization of Bio-Oil
via Hydrotreating



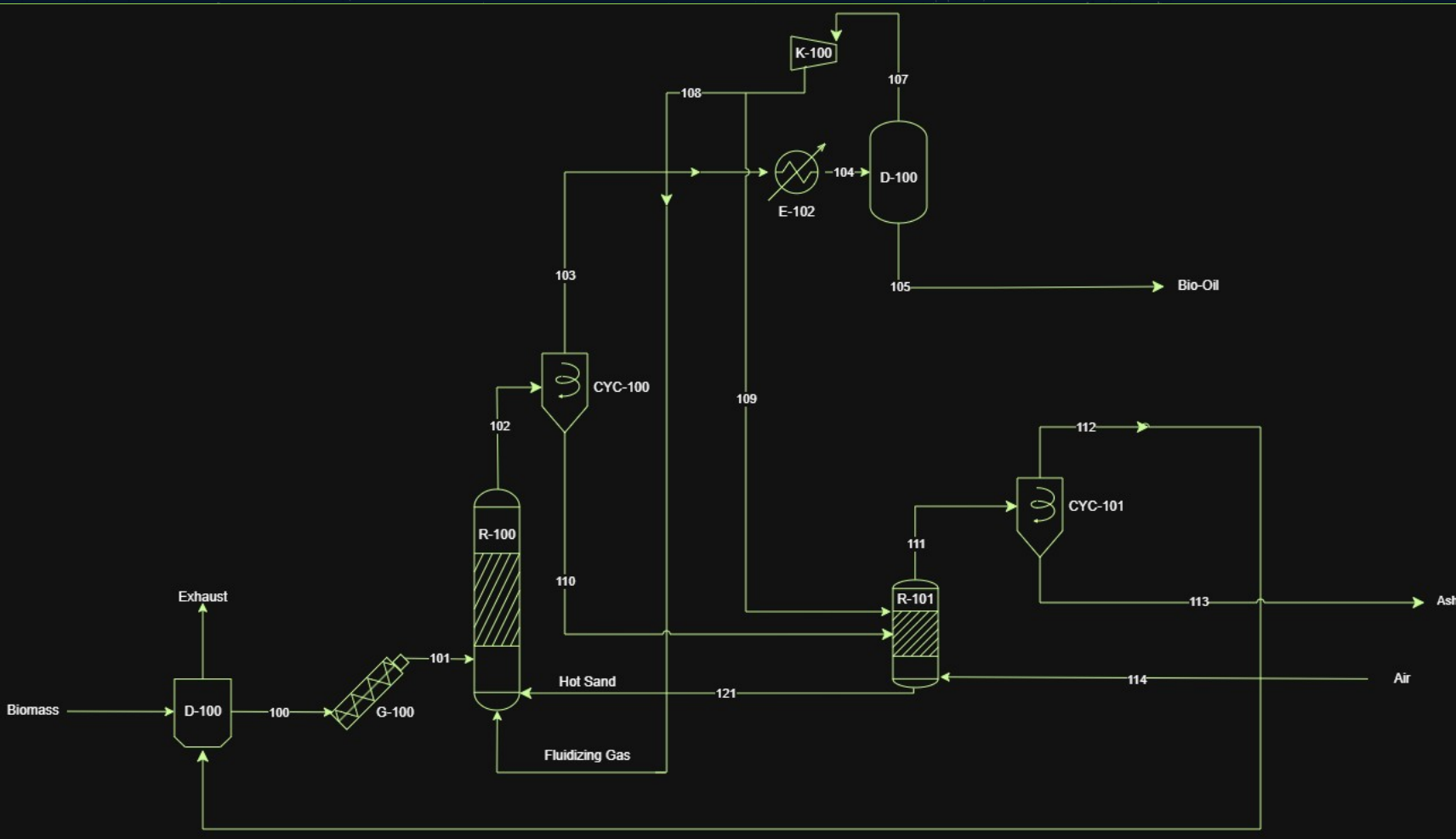
Fractionation

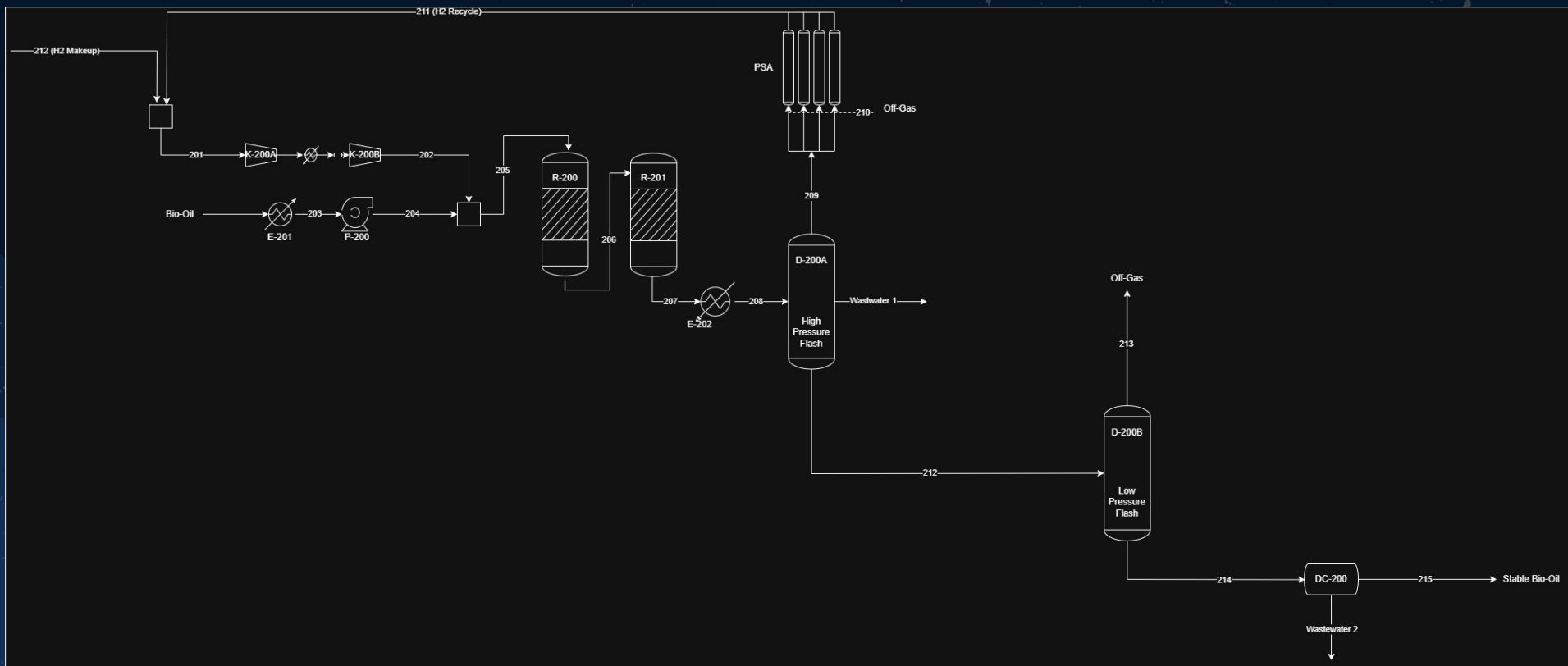
Debutanization and
Fractionation into
RP-1

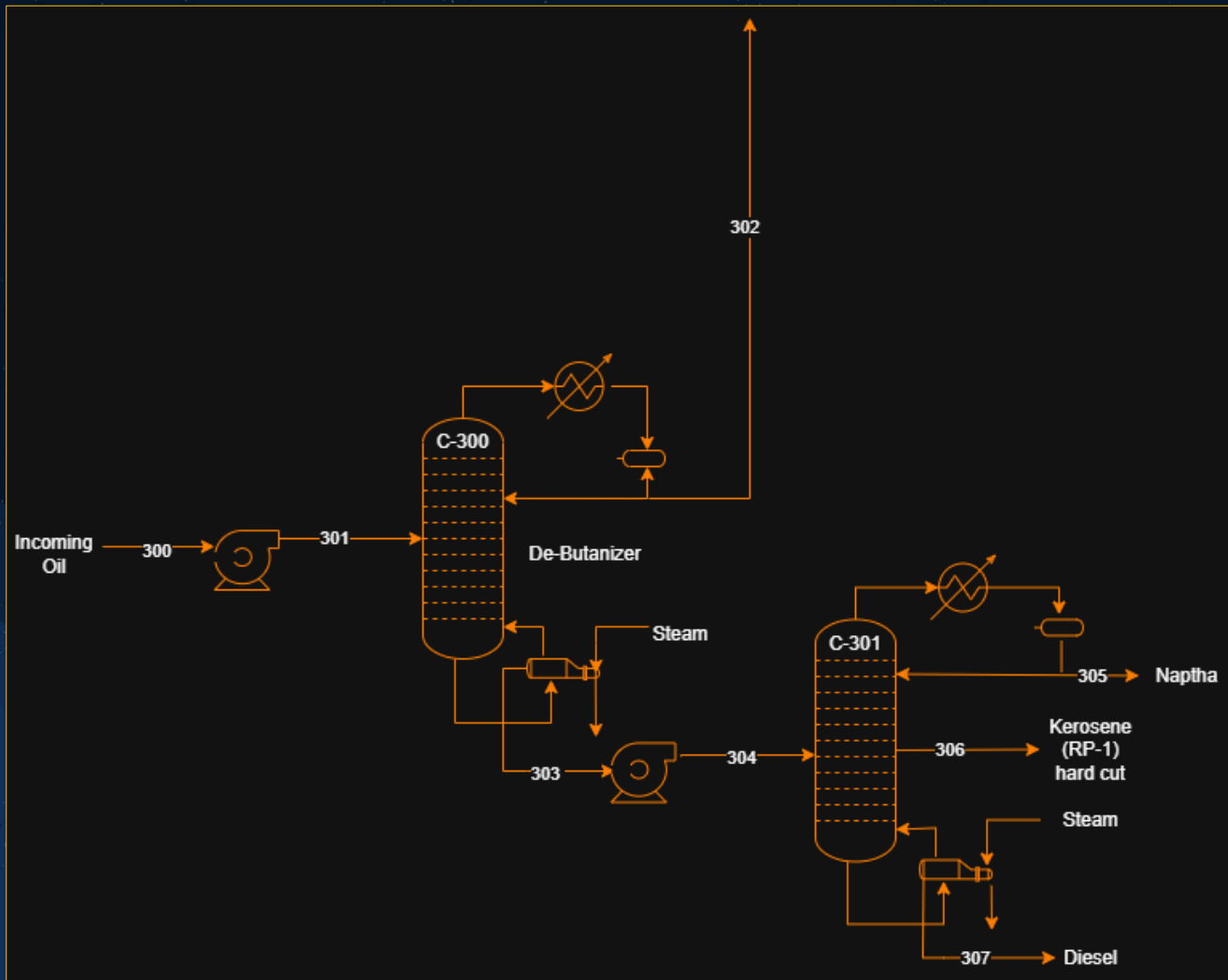




PFD



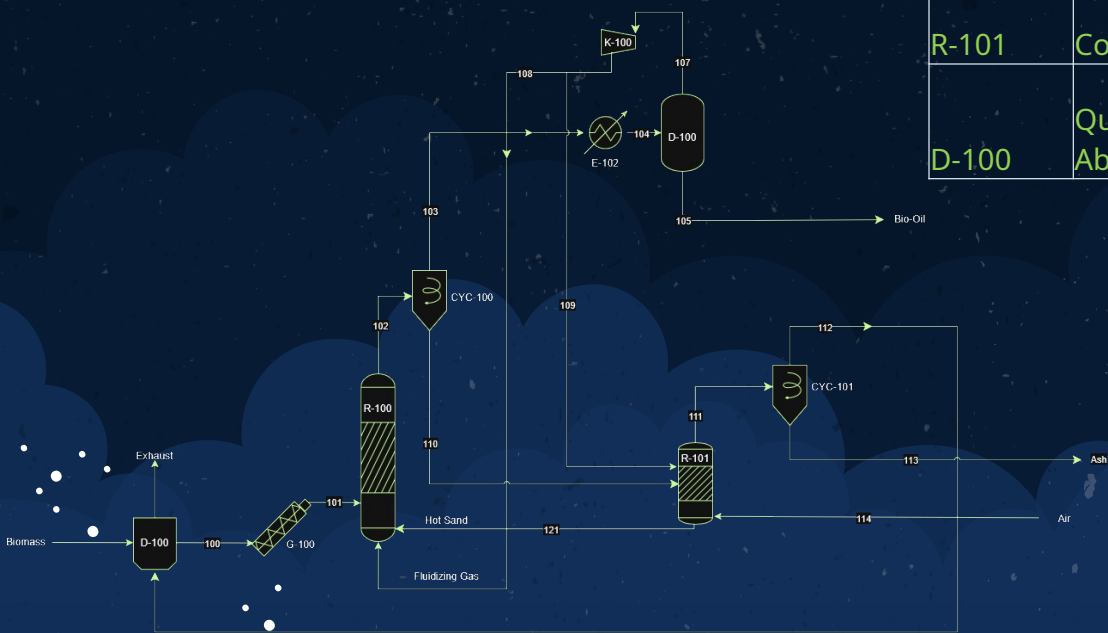




The background is a dark blue space scene. It features several white stars of different sizes, a planet with light blue and white horizontal stripes in the upper right, and a large starburst with radiating lines in the upper left. The bottom of the image is filled with a pattern of overlapping dark blue circles, resembling a nebula or a field of distant galaxies.

Equipment

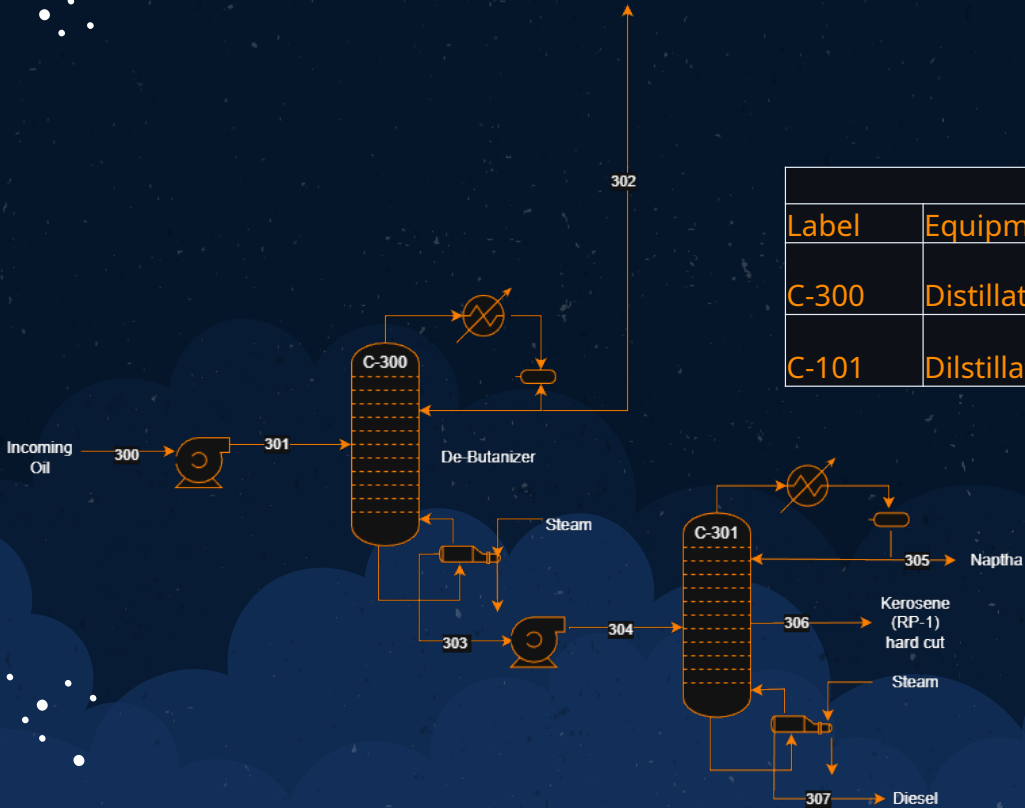
Fast Pyrolysis



Fast Pyrolysis			
Label	Equipment	Material	Sizing
D-100	Dryer	Carbon Steel	Tray Area = 200.21 sq ft
R-100	Pyrolysis Rise Reactor	316 SS	Diameter = 0.98 in Length = 20.01 ft
R-101	Combustor	316 SS	Volume = 634.52 US gal Diameter = 2.99 ft Height = 12.01 ft
D-100	Quench Absorber	Carbon Steel	Diameter = 7 ft Height = 17.06 ft



Fractionation



Fractionation			
Label	Equipment	Material	Sizing
C-300	Distillation Columns	Killed Carbon Steel	Diameter = 4 ft Height = 121.4 ft
C-101	Dilstillation Columns	Killed Carbon Steel	Diameter = 4 ft Height = 58.1 ft

Preliminary Utilities

Preliminary Utilities Summary		
Type	Cost	Rate
Low Pressure Steam (\$/kWh)	\$ 0.022	2000 kWh
Cooling Water (\$/gal)	\$ 0.002	45,000 gal/hr
Electricity (\$/kWh)	\$ 0.06	1064.79 kWh
Wastewater (\$/gal)	\$ 0.002	10,097 gal/hr
Fired Heat (\$/kWh)	\$ 0.02	22,567 kWh



Safety Analysis and Process Control Loop

Safety Analysis

Component

Safety + Exposure Limits

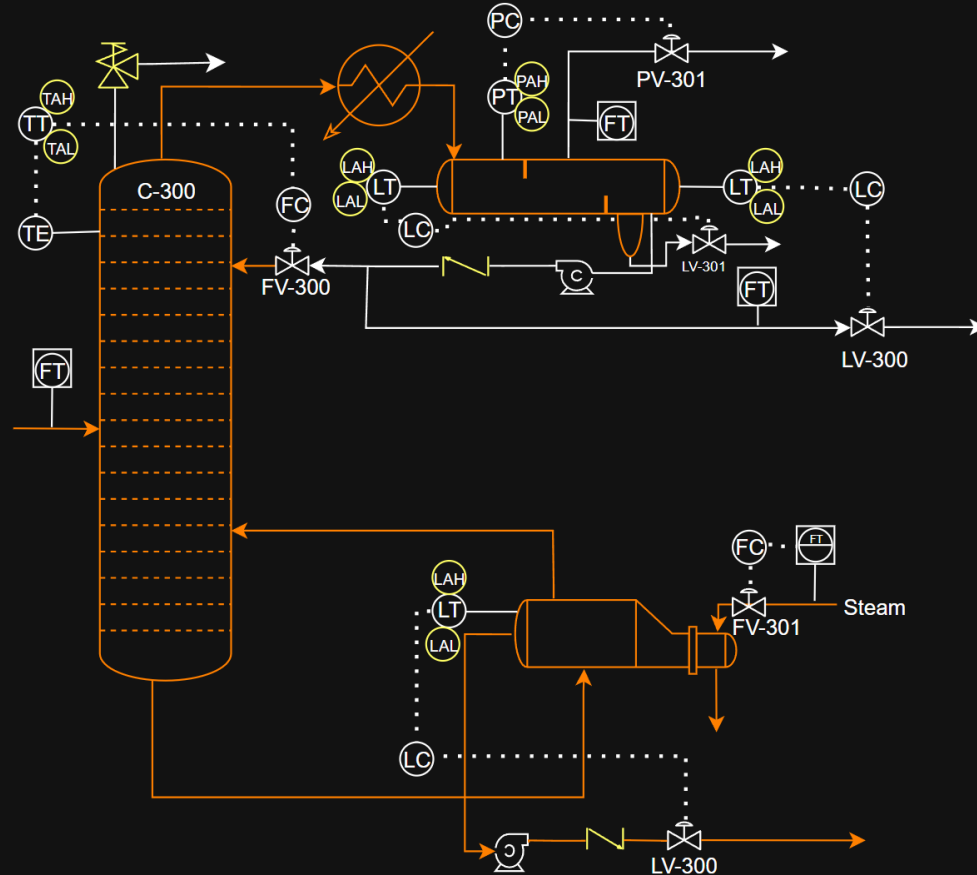
Hydrogen	1. Highly flammable gas 2. OSHA Hazardous Communication Standard 3. Exposure Limit: Simple asphyxiant 4. Hydrogen flames may be nearly invisible in daylight
Carbon Monoxide	1. Toxic and flammable gas 2. OSHA Hazardous Communication Standard 3. Exposure Limit: TWA: 50 ppm over an 8-hour shift
Methane	1. Highly flammable gas 2. Explosion hazard in confined spaces 3. Exposure Limit: Simple asphyxiant
Ethylene	1. Flammable gas 2. Explosion risk in process equipment 3. Exposure Limit: Simple asphyxiant
Sulfur Dioxide	1. Toxic respiratory irritant 2. Corrosive gas 3. Exposure Limit: 5 ppm (OSHA ceiling limit)
Nitrogen Dioxide	1. Toxic oxidizing gas 2. Lung irritant 3. Exposure Limit: 5 ppm (OSHA ceiling limit)



Safety Analysis

Component	Safety + Exposure Limits
Acetone	1. Highly flammable liquid 2. OSHA Hazardous Communication Standard 3. Exposure Limit: TWA: 1000 ppm over an 8-hour shift
Acetic Acid	1. Corrosive liquid 2. Skin and respiratory irritant 3. Exposure Limit: TWA 10 ppm (OSHA PEL)
Furfural	1. Toxic aldehyde 2. Flammable liquid 3. Exposure Limit: TWA: 2 ppm over an 8-hour shift
Carbon Dioxide	1. OSHA Hazardous Communication Standard, 2. Simple asphyxiant 3. Exposure Limit: TWA 5000 ppm (OSHA PEL)

Process Control on Debutanizer





Economic Analysis

Cost Breakdown

Plant Life	Credits	Product Pricing	Initial Investment
California Green Refinery	Low Credit Fuel Standard (LCFS)	RP-1, Biodiesel (B99-B100), Naphtha	MACRS-7
21 years	\$0.75/gal	RP-1: \$3.0/lb Diesel: \$2.3/gal Naphtha: \$1/gal	25%

Capital Cost Summary

ISBL & OSBL

ISBL: \$103MM

OSBL: \$41.5MM

Working Capital

WC: \$15.6MM

ECC & Contingency

ECC: \$10.4MM

Contingency: \$10.4MM

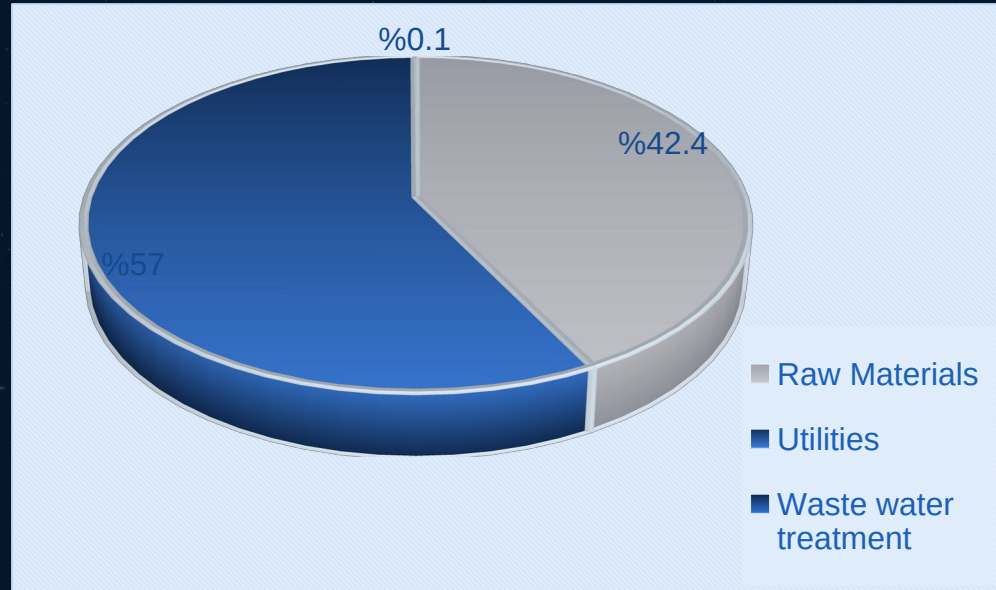
TCC & FCC

TCC: \$181.7MM

FCC: \$145.4MM

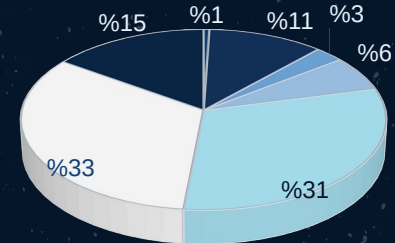
Economic Analysis: VCOP

VCOP Sumary (\$MM/yr)	
Raw Materials	79.2
Utilities	107.58
Waste water treatment	0.161
Total VCOP	186.94



ECONOMIC ANALYSIS: FCOP

FCOP	
Number of shifts per day	4
Number of operators per shift	5
Operator salary(\$/yr)	120,000
Total operator Salary(\$/yr)	2,400,000
Supervisor 25% of operator	600,000
Direct Salary Overhead 45% of Labor and Superv) \$/yr	1,350,000
Maintenance 3% of ISBL	6,480,000
Plant Overhead 65% of Labor and Maintance	7,039,500
Tax and Insurance 2% of fixed Investment	3,240,000
FCOP	21,109,500



ECONOMIC ANALYSIS: REVENUE

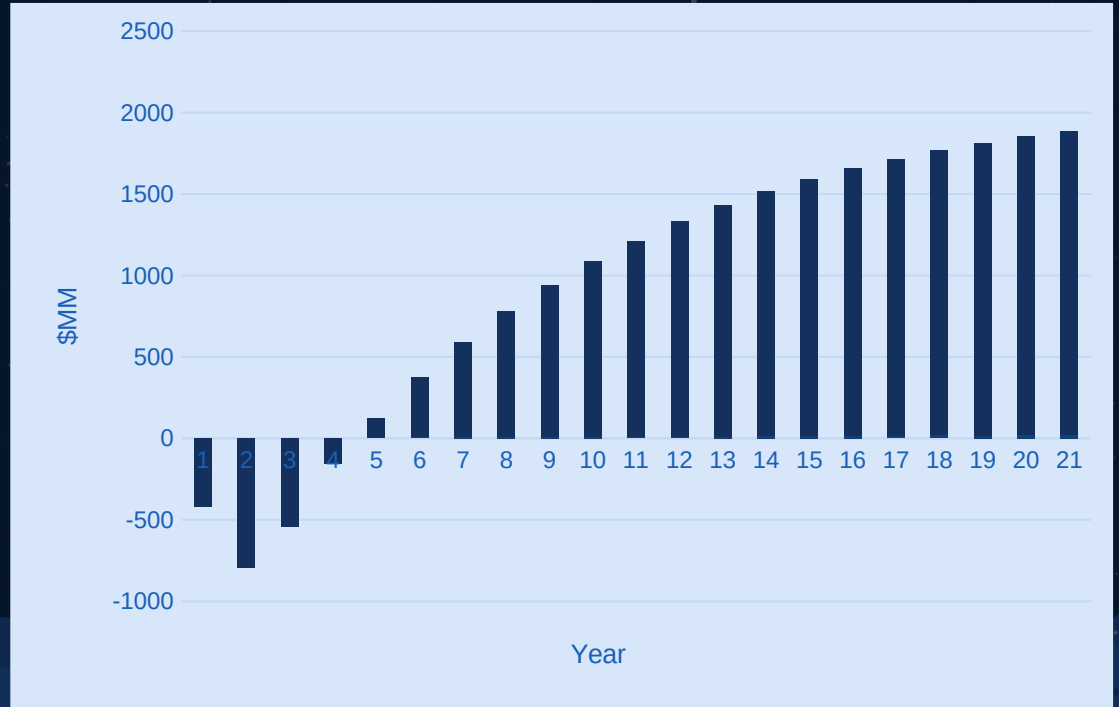
Product	Price \$/unit	Rate	Profit \$MM/yr
RP-1	3.00	31858.7 lb/hr	764.6
Diesel	2.3	8489.62 gal/hr	156.2
Naphtha	1.00	5411.69 gal/hr	43.3

NET PRESENT VALUE

NPV (7 year MACRS)

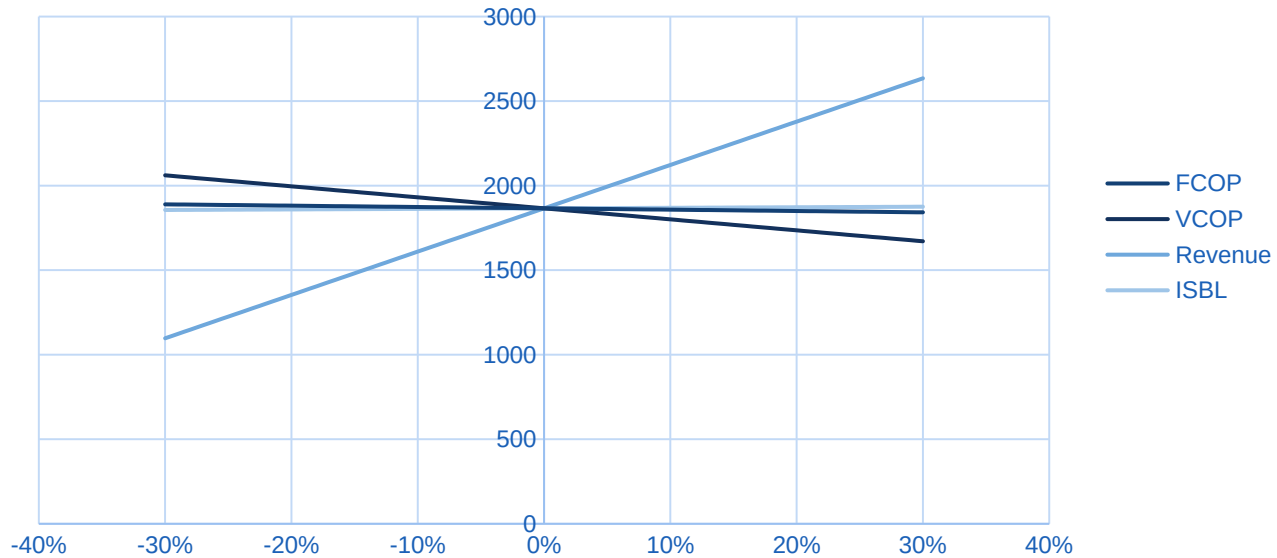
Gross Profit: \$1865MM

Plant life : 21 years



SENSITIVITY ANALYSIS

%Change VS NPV for 7 year MACRS



Thank You!

